

SENTINEL XDR: An LLM-Augmented, Multi-Stage Intrusion Detection and Response System with Retrieval-Grounded Reasoning

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Abstract—Classical intrusion detection systems (IDS) are precise for known signatures but blind to novel, multi-stage attacks, while machine-learning detectors generalise yet emit uninterpretable scores. We present SENTINEL XDR, a defensive AI-IDS that unifies a deterministic heuristic triage stage with a retrieval-augmented Large-Language-Model (LLM) analyst and a multi-stage correlation engine. Live packets are captured with Scapy, screened by ten heuristic rules over an amortised $O(1)$ sliding window, and only suspicious events are escalated to an LLM (NVIDIA NIM with Gemini/Groq fallback) whose prompt is grounded with MITRE ATT&CK knowledge retrieved from a ChromaDB vector store. The model returns a structured, human-readable verdict—threat level, attack vector, MITRE technique, confidence, and recommended action. A correlation engine (MACE) links atomic alerts into kill-chain attack chains, with companion engines for attacker profiling, dry-run response, executive narratives, and prompt-injection defence. On synthetic multi-protocol attack traffic the system classified all seven evaluated attack classes (100% coverage), and on live Wi-Fi capture it sustained up to ~ 390 packets/s at constant per-packet triage cost. The result is an explainable, grounded, and resilient detection-and-response pipeline targeting the interpretability, grounding, and multi-stage gaps of prior work.

Index Terms—intrusion detection, large language models, retrieval-augmented generation, MITRE ATT&CK, XDR, network security, SOC automation.

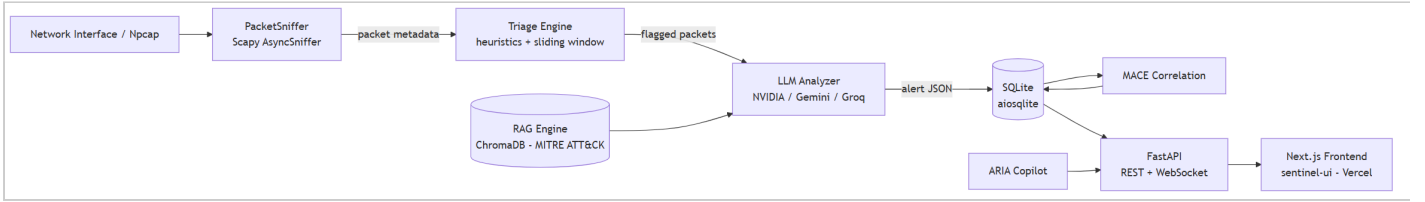


Fig. 1. SENTINEL XDR system architecture: capture $\rightarrow$ heuristic triage $\rightarrow$ RAG-grounded LLM analysis $\rightarrow$ persistence $\rightarrow$ MACE correlation and API/UI.

I. INTRODUCTION

SOC analysts are overwhelmed by high alert volumes and low-context output, producing alert fatigue and slow response. Signature systems such as Snort [1] and Suricata [2] are precise but reactive and campaign-unaware. Machine-learning and anomaly detectors [3],[4],[10] generalise but output opaque labels or scores—an interpretability gap repeatedly cited as the central operational obstacle [4]. Recent LLM-augmented tools [7],[8] promise readable analysis yet are typically ungrounded and detached from a live capture pipeline.

This paper contributes SENTINEL XDR, a single pipeline combining (i) fast deterministic triage, (ii) a retrieval-grounded multi-provider LLM analyst returning explainable, MITRE-mapped verdicts, (iii) multi-stage kill-chain correlation, and (iv) safe dry-run response and reporting. Objectives: capture live traffic and flag suspicion efficiently; escalate only suspicious events to an LLM grounded via RAG [6]; correlate alerts into chains; and deliver an analyst-grade interface with an honest cloud Demo Mode.

II. RELATED WORK

Snort [1] established rule-based detection; Suricata [2] added multi-threading and flow/TLS awareness, but both remain signature-bound. CICIDS2017 [3] and ML-IDS surveys [4] advanced classification while leaving interpretability unsolved. Kitsune [10] enabled unsupervised online anomaly detection without labels, yet scores are not human-readable. MITRE ATT&CK [5] standardises adversary description but is not a detector. RAG [6] grounds LLM generation in an external corpus. Surveys of generative AI in security [7] and emerging LLM-IDS work [8] show promise but are generally ungrounded and offline. Table I positions SENTINEL against these works.

Table I. Capability comparison.

Capability	Snort/Suri.	ML-IDS	LLM	Ours
Known-attack detection	✓	✓	—	✓
Novel generalisation	—	part	✓	✓
NL explanation	—	—	✓	✓
MITRE per alert	part	—	part	✓
Multi-stage correlation	—	—	—	✓
RAG grounding	—	—	part	✓
Live capture	✓	var	—	✓
Dry-run response	—	—	—	✓

III. SYSTEM DESIGN

SENTINEL is a five-stage pipeline—Capture, Triage, AI Analysis, Persist, Correlate/Serve—surrounded by an XDR engine suite (MACE, ARIA, ADRS, PHANTOM, AEGIS, CHRONICLE). Fig. 2 shows the detection pipeline.

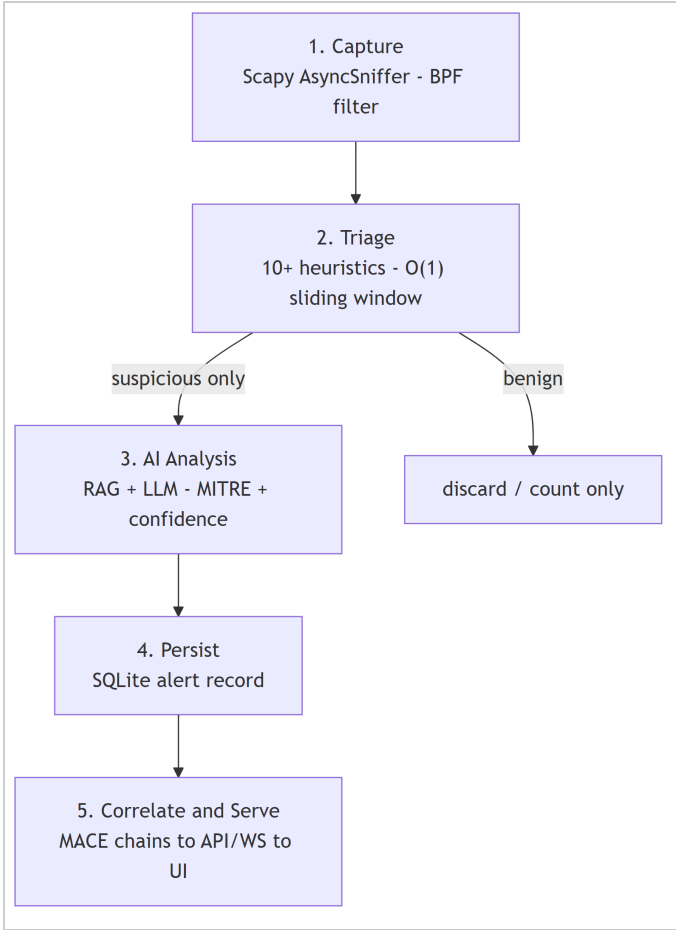


Fig. 2. Five-stage detection pipeline; only flagged packets reach the LLM.

A. Threat Model

Applying STRIDE with the IDS as the asset: capture stores metadata only (information disclosure); bounded asynchronous queues with back-pressure mitigate denial of service; capture is gated behind explicit operator action and disabled in cloud (elevation of privilege). For the LLM surface, AEGIS scans inputs for prompt injection and provider keys remain backend-only secrets.

IV. IMPLEMENTATION

Capture & Triage. PacketSniffer wraps Scapy's AsyncSniffer with a BPF filter and extracts a 5-tuple, TCP flags, size, and bounded payload-hex. Ten heuristics (SYN/port/NULL/XMAS/FIN scans, ICMP flood, DNS tunnelling, large payload, suspicious ports, high-frequency) run over a thread-safe sliding window at amortised $O(1)$ per packet; only flagged packets proceed.

Grounded LLM Analysis. For each flagged packet, relevant MITRE techniques are retrieved from ChromaDB and injected into a SOC-analyst prompt. Providers fail over NVIDIA → Gemini → Groq with exponential back-off; the response is parsed into a structured alert. On total failure the system degrades to a heuristic-only alert.

Correlation. MACE matches alert sequences against kill-chain templates to build scored attack chains; CHRONICLE generates executive narratives; PHANTOM profiles actors; ADRS proposes dry-run policies; ARIA streams a RAG-grounded conversation. The backend is FastAPI (REST + WebSocket, aiosqlite); the frontend is a Next.js command-center. Deployment: frontend on Vercel, backend on a Hugging Face Docker Space, with Demo Mode where raw capture is unavailable.

V. EXPERIMENTAL EVALUATION

Experiments ran on a Windows 11 host with Npcap over Wi-Fi; the primary LLM was NVIDIA NIM (Llama-3.3-70B). Detection was validated by injecting labelled attacks via Demo Mode and recording classifications; performance was measured via the status/statistics endpoints.

Table II. Attack-class detection (sampled).

Attack class	Heuristic → MITRE	Det.	Ev.
SYN scan/flood	SYN_SCAN → T1595/T1499	✓	62
Port scan/sweep	PORT_SWEEP → T1046	✓	57
ICMP flood	ICMP_FLOOD → T1499	✓	61
NULL scan	NULL_SCAN → T1595	✓	62
XMAS scan	XMAS_SCAN → T1595	✓	59
Suspicious port 4444	SUSPICIOUS_PORT	✓	56
DNS tunnelling	DNS_TUNNEL → T1048	✓	51

All seven evaluated classes were detected (100% coverage). Severity over 777 events: Critical 51, High 118, Medium 536, Low 72; protocols UDP 577, TCP 563, ICMP 61.

Table III. Measured performance.

Metric	Measured
Capture (idle)	8.7 pkt/s (384/44 s)
Capture (active)	~390 pkt/s (~6,222/16 s)
Triage complexity	$O(1)$ amortised/packet
LLM throughput	0.50 ev/s (~2.0 s/ev)
Detection coverage	7/7 (100%)

Live capture sustained 8.7 pkt/s idle and ~390 pkt/s under load at constant triage cost; the LLM stage (~2 s/event) is the bottleneck, mitigated by escalating only the flagged subset. Benign traffic produced no critical detections in test windows.

VI. DISCUSSION

Separating fast deterministic triage from slow semantic analysis bounds per-packet cost at $O(1)$ while reserving expensive reasoning for the suspicious minority. RAG grounding keeps MITRE references accurate and explanations actionable, addressing interpretability and hallucination gaps. Limitations: LLM latency under sustained load and metadata-only visibility of encrypted C2.

VII. CONCLUSION AND FUTURE WORK

SENTINEL XDR shows that a hybrid heuristic + RAG-grounded LLM + multi-stage correlation architecture yields explainable, MITRE-mapped, campaign-aware detection beyond classical and ML IDS, with a production-grade implementation and cloud deployment. Future work: backend authentication and rate-limiting, WebSocket real-time alerting, persistence of analyst actions/reports, a Local Authorized Agent for audited response, a fine-tuned local model, and formal CICIDS2017 evaluation.

REFERENCES

- [1] M. Roesch, "Snort—Lightweight Intrusion Detection for Networks," Proc. 13th USENIX LISA, 1999, pp. 229–238.
- [2] OISF, "Suricata IDS/IPS Engine," 2010. [Online]. <https://suricata.io>
- [3] I. Sharafaldin, A. H. Lashkari, A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset...," Proc. ICISP, 2018, pp. 108–116.
- [4] A. L. Buczak, E. Guven, "A Survey of Data Mining and ML Methods for Cyber Security Intrusion Detection," IEEE Commun. Surveys Tuts., 18(2):1153–1176, 2016.
- [5] B. E. Strom et al., "MITRE ATT&CK: Design and Philosophy," MITRE Corp., 2018.

- [6] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Proc. NeurIPS, 2020.
- [7] M. Gupta et al., "From ChatGPT to ThreatGPT...", IEEE Access, 11:80218–80245, 2023.
- [8] M. A. Ferrag et al., "Generative AI and LLMs for Cyber Security: A Review," 2024.
- [9] A. Vaswani et al., "Attention Is All You Need," Proc. NeurIPS, 2017.
- [10] Y. Mirsky et al., "Kitsune: An Ensemble of Autoencoders for Online Network Intrusion Detection," Proc. NDSS, 2018.